

BIOMEDICAL IMAGE ANALYSIS PROJECT SS2026

Kickoff Meeting

Mental model and BIMAP aims

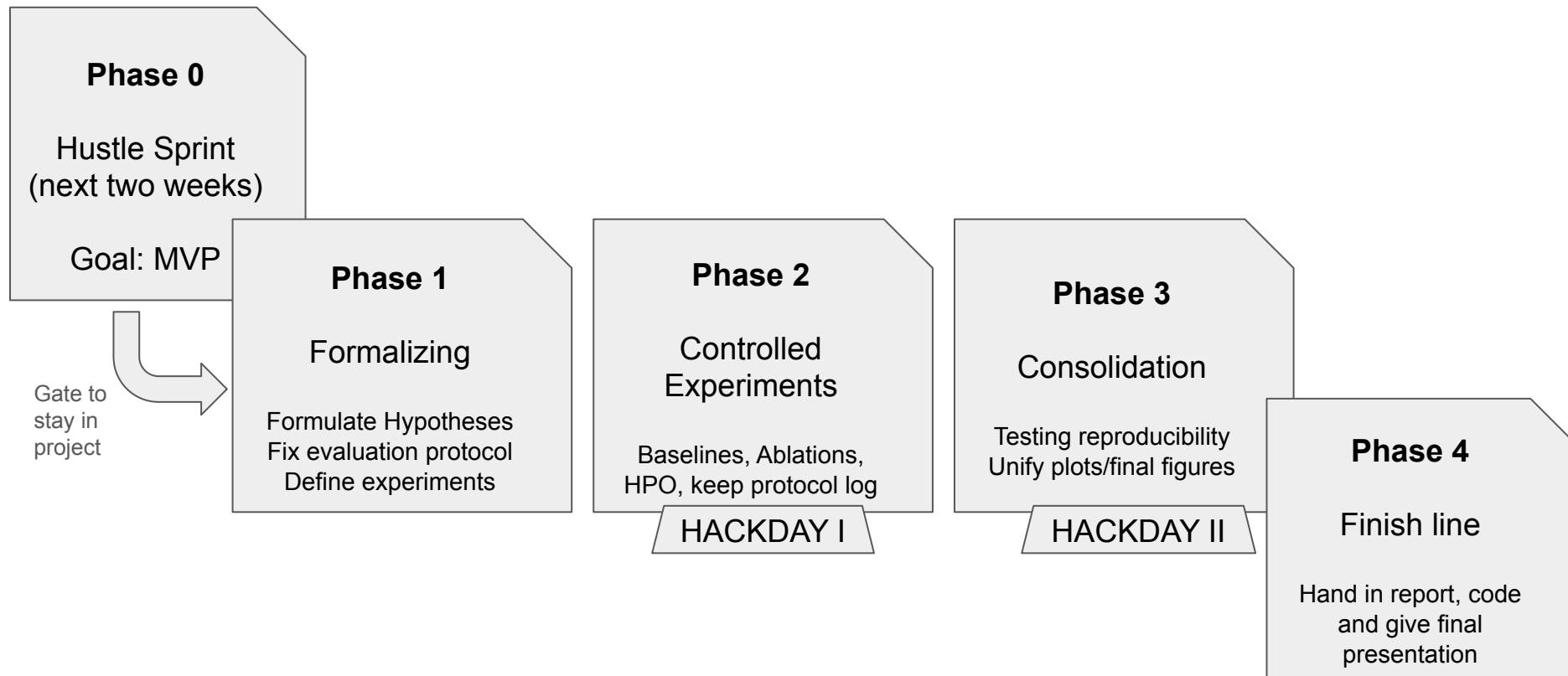
You will learn to think like a researcher, not how to become a prompt engineer!

Your project should look like a small NeurIPS paper, not a Kaggle notebook.

Projects are designed being fun and ambitious and they need your time (10 ECTS ~ 300 working hours)

- Many students continued project as MSc thesis in the lab
- People wrote or write papers about their BIMAP project
 - Resource-efficient training and inference of TransUNets (BIMAP 2025, in preparation)
 - Neuro-SAM - segmentation foundation models for neuroscience (BIMAP 2024, in prep)
 - Rodríguez, B. L., Borges, G. L., García, M. L. B., Schützenberger, A., & Kist, A. M. (2024, May). Neural Radiance Fields for 3D Reconstruction in Monoscopic Laryngeal Endoscopy. In 2024 IEEE International Symposium on Biomedical Imaging (ISBI) - BIMAP 2023

Seminar overview



Timetable



A	B	C	D	E	F
Meetings BIMAP General Assembly					
14.04.2026	Tuesday	16.00-17.30	Kickoff	Explaining course and pitch projects	
Phase 0: Hacking time, preparing MVP and presentation					
24.04.2026	Friday	12.15-13.45	MVP Presentations	Phase 0 achievers stay	
Phase 1: Formulate Hypotheses, fix evaluation protocol, define experiments					
08.05.2026	Friday	12.15-13.45	Phase 1 - Interim presentation		
Phase 1 Reiteration					
22.05.2026	Friday	12.15-13.45	Phase 1 - Interim presentation, updates		
Phase 2: Controlled Experiments					
29.05.2026	Friday	12.15-13.45	Phase 2 - Interim report		
Phase 2: Controlled Experiments					
12.06.2026	Friday	14.00 - 18.30	Hackday I (Dinner at 18.30)		
Phase 2: Controlled Experiments					
26.06.2026	Friday	12.15-13.45	Phase 2 - Interim report		
Phase 3: Consolidation					
03.07.2026	Friday	14.00 - 18.30	Hackday II (Dinner at 18.30)		
Phase 4: Finalizing					
14.07.2026	Tuesday	16.00-17.30	Phase 4 - Interim report, wrapping up		
Presentations: 23.07.2026		09.00-13.00			
Report due: 31.07.2026					

Alternative: July 2nd
June 29th

Science is key.

We enforce scientific methods.

This is not how it works:

“We tried many things.”

“After tuning...”

“We experimented with different architectures.”

“We achieved up to...”

Rather

“We evaluated the following controlled variants.”

“We tested X while holding Y constant.”

Every experiment must have:

1. Hypothesis
2. Single controlled change
3. Documented configuration
4. Run ID
5. Metric
6. Interpretation

Compute

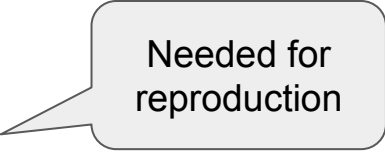
You can use Google Colab, our workstations or the HPC.
We recommend using the HPC.

(let me know if you need access, such that I can give you this)



https://doc.nhr.fau.de/getting_started/

- Attendance is mandatory
(you can miss once, twice with a doctor's note) → otherwise out
- Intermediate presentations
(not graded, but have to be given, if sick you need alternative dates, we make your life easy and provide templates to be filled by you)
- Get things done (hacking day I)
- Reproducing code from other projects (hacking day II)
- For grade:
 - Prototype/code
 - Presentation (15 min)
 - Report (15 pages)

A light gray speech bubble with a black border and a tail pointing towards the first bullet point. It contains the text 'Needed for reproduction' in a black sans-serif font.

Needed for
reproduction

- Version control (Github) is mandatory
 - Experiment log is mandatory
 - Documentation and comments are mandatory
-
- Unit tests depending on project
 - Tutorial for newcomers depending on project
 - Python package pip installable depending on project

- 15 pages in JMLR template
- Report has to be written using LaTeX (Overleaf project)
- Content like a scientific paper
 - Abstract
 - Introduction
 - Methods
 - Results
 - Discussion
 - References

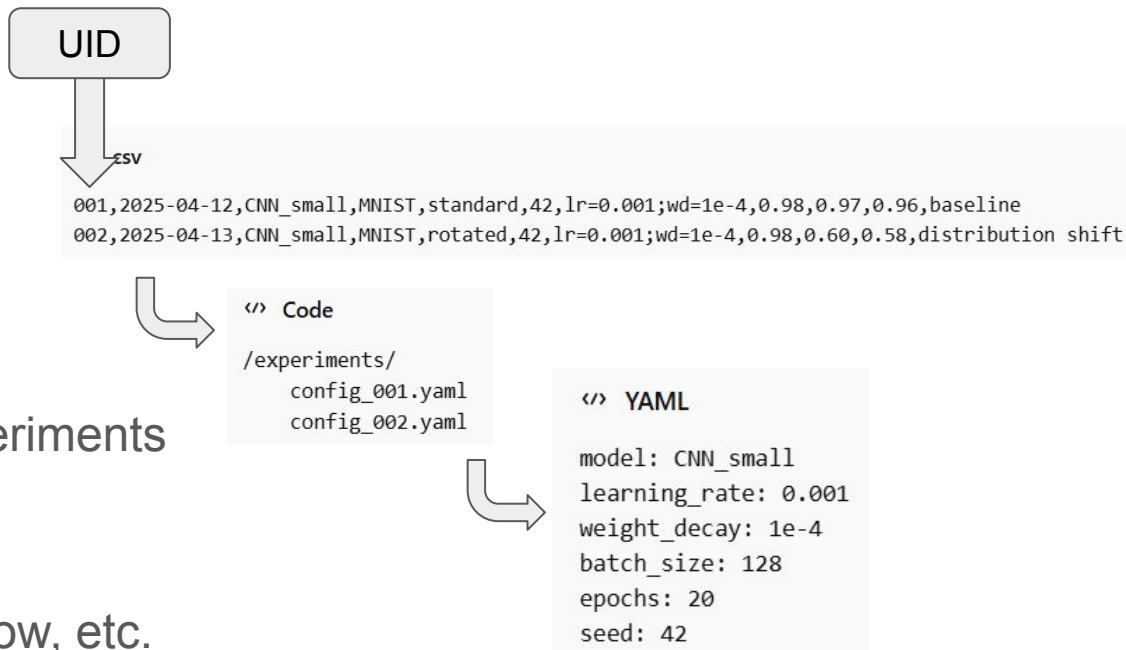
Experimental log is mandatory.

Tier A (mandatory)

- Experiments.csv
- Config.yaml per run
- 3 seeds minimum per experiments

Tier B (optional)

- Use tools like wandb, MLflow, etc.



Main deliverables and grading

Code

Reproducibility (5P)
Structure and
Documentation (5P)
Evaluation correct (5P)
Experimental log (5P)

10P = 4.0

Report

Problem clear (5P)
Methods clear (5P)
Evaluation correct (5P)
Critical discussion (5P)

10P = 4.0

Presentation

Clarity (5P)
Technical depth (5P)
Reflexion (5P)
Q&A (5P)

10P = 4.0

Final grade = (Code + Report + Presentation) / 3

We will have two afternoons where we start at **2 PM** and hack until dinner at **6.30 PM**. We will provide dinner (pizza or similar, vegetarian or vegan possible).

The hacking days will have a clear structure.

Aim for today (will be
written on board)

Hacking

Results presentation
(very short)

Dinner

More hacking or depart

- It is not forbidden, but explicitly discouraged
- You need to follow the scientific principle and not vague descriptions
- The team will do code reviews with you - you need to be able to explain each single line of code (*high level, e.g. what does `np.dot` exactly here - e.g. computing the local attention mask, not how `np.dot` works under the hood*)

Vibe	Science
"We improved it."	"Run 14–16 show $+3.2 \pm 0.4\%$ improvement."
"We tuned hyperparameters."	"We tested $lr \in \{1e-3, 1e-4\}$ with seed $\{1, 2, 3\}$."
"It works better."	"Statistically significant at $p < 0.05$."

Projects

TL;DR: The project examines powerful biomedical image segmentation versions of very small ($<0.1\text{M}$ params) U-Nets and tests their generalization performance.

Narrow and on the edge.

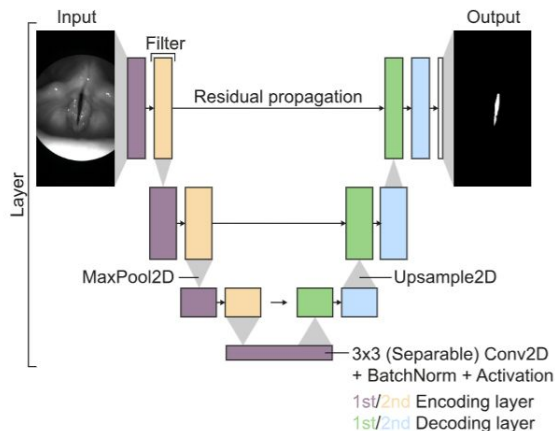


FIGURE 2. Our U-Net baseline variant is based on an encoder/decoder architecture that contains two convolutional layers in each main layer, followed by a MaxPool2D and Upsample2D operation in the encoder and decoder, respectively. Residuals are propagated from encoder to decoder by default via concatenation.

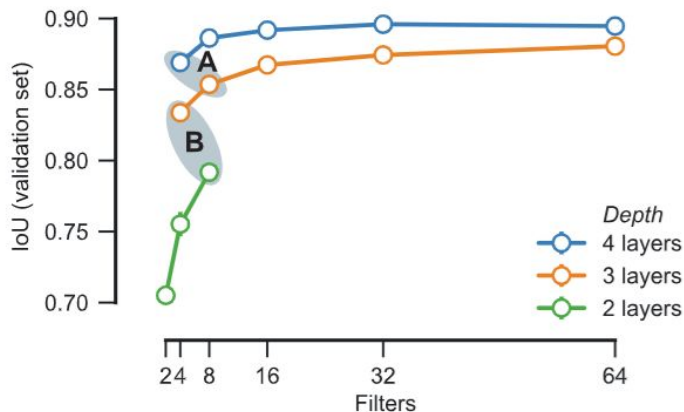
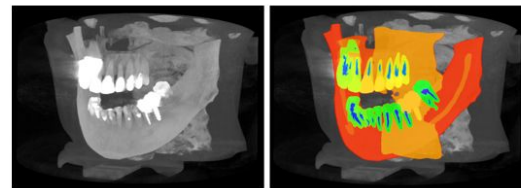
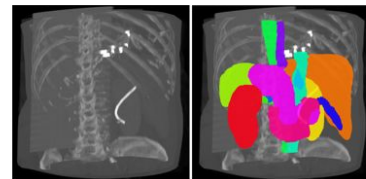
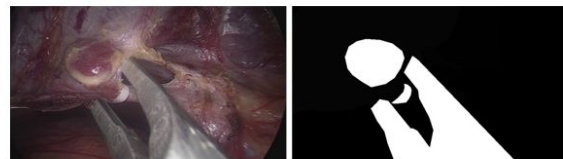
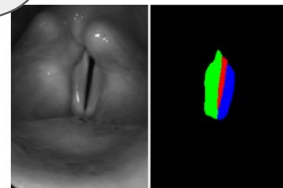
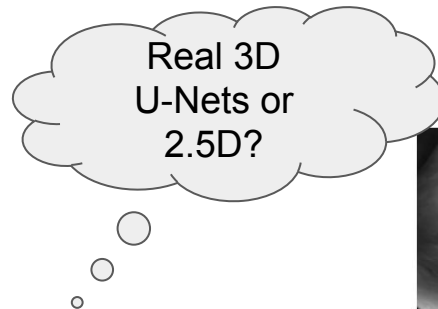
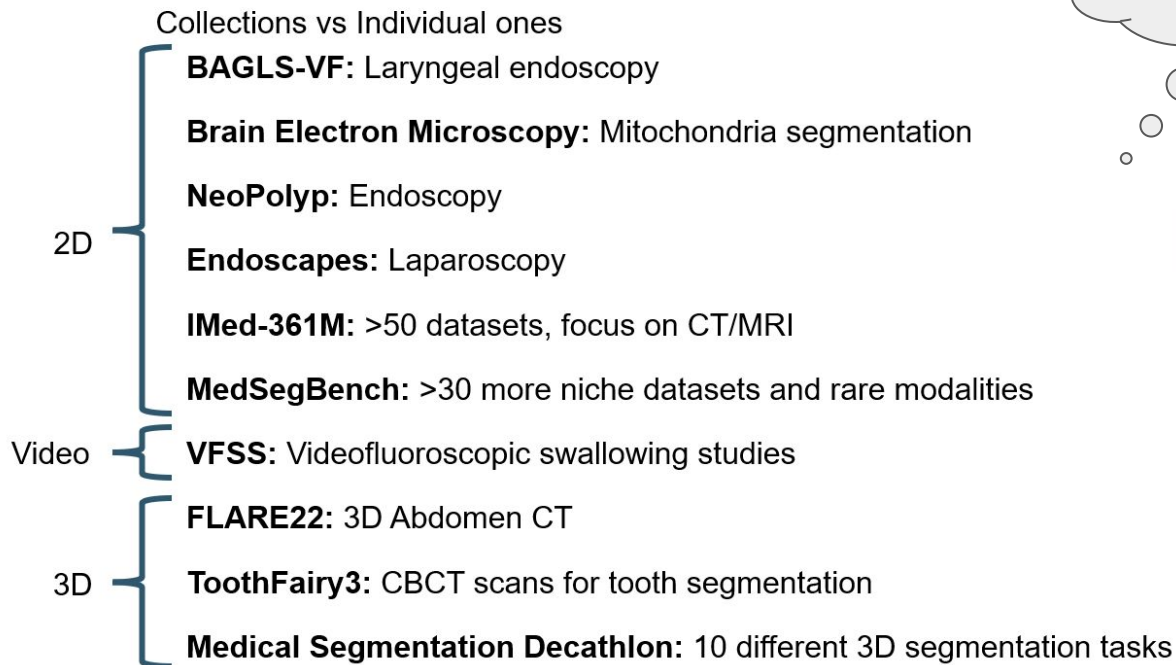


FIGURE 3. Segmentation accuracy is stable across most layer and filter settings. IoU: Intersection over Union. Gray ellipses (noted A and B) indicate two settings with almost identical number of parameters.

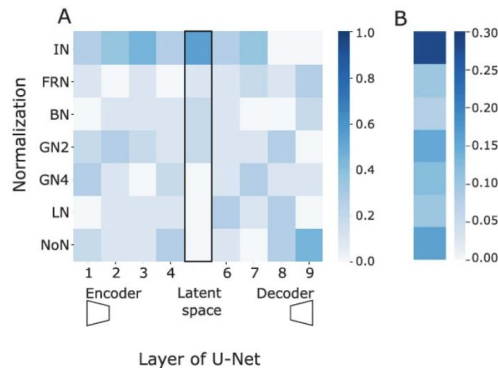
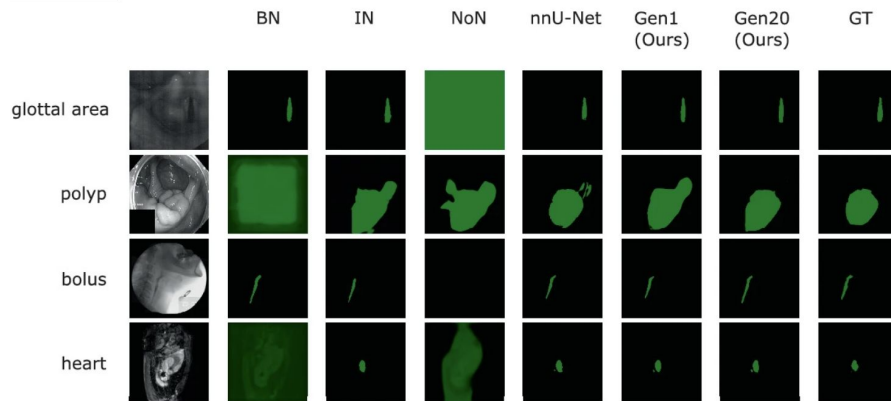
Kist and Döllinger,
IEEE Access 2020

Datasets to be considered



What can we investigate?

- Develop multiple U-Net architectures $<0.1\text{M}$ parameters using different normalization patterns, attention schemes, residual propagation, depth, etc.
- neural architecture search
- quantization



Neubig and Kist,
MICCAI 2023

Overall goal

The goal of this project is to investigate the relationship between model size and performance systematically by analyzing the trade-off between parameters, compute and accuracy across datasets and U-Net implementations.

Minimal viable product (Phase 0 deliverable):

By the end of Phase 0, the project should produce a working end-to-end prototype. This includes loading the BAGLS dataset, implementing one U-Net architecture, training the model, evaluating its segmentation performance on a validation set, and logging all experiments in a structured way. The baseline model should have at most 0.1M parameters.

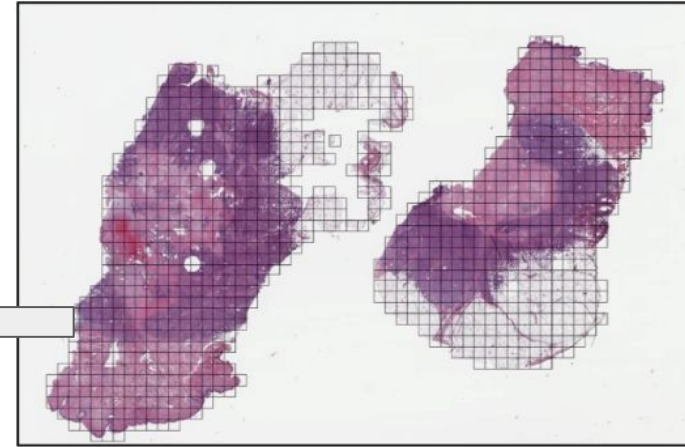
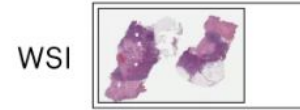
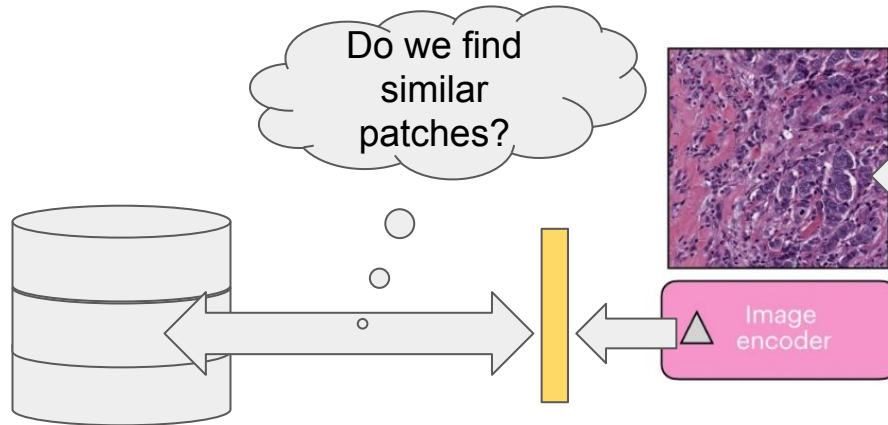
Assigned to **MICROUNET**:

Bhavan Kumar Velladurai

Jakir Hussain Rifat

Shivangi Jitendrabhai Ratnu

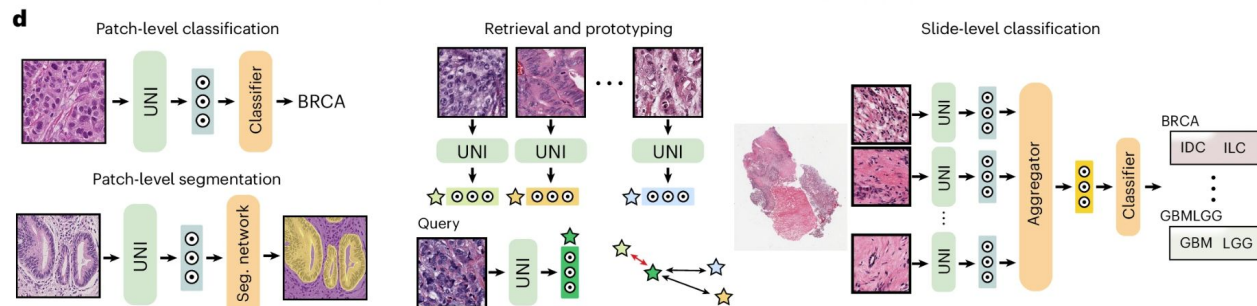
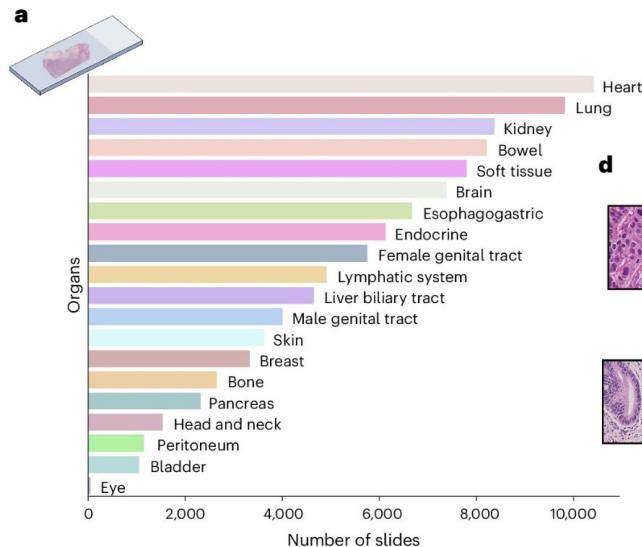
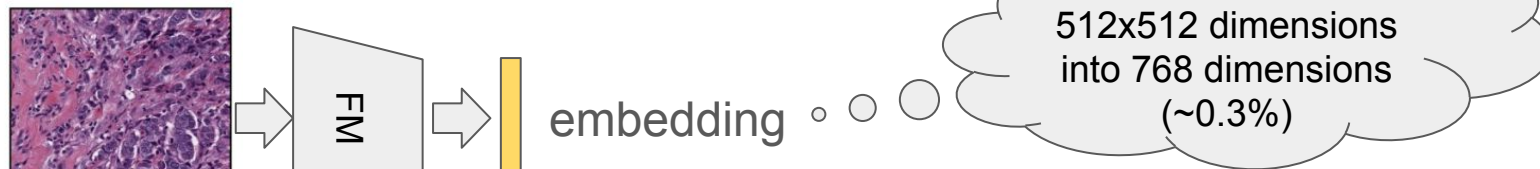
TL;DR: This project investigates whether retrieval-augmented methods can be used to build an interactive histopathology atlas from whole slide images (WSIs). Image patches extracted from slides are embedded using pretrained vision or vision-language models and stored in a vector database. The system should allow image or text queries to retrieve visually similar tissue regions and evaluate retrieval performance on labeled histopathology data.



Tiled WSI

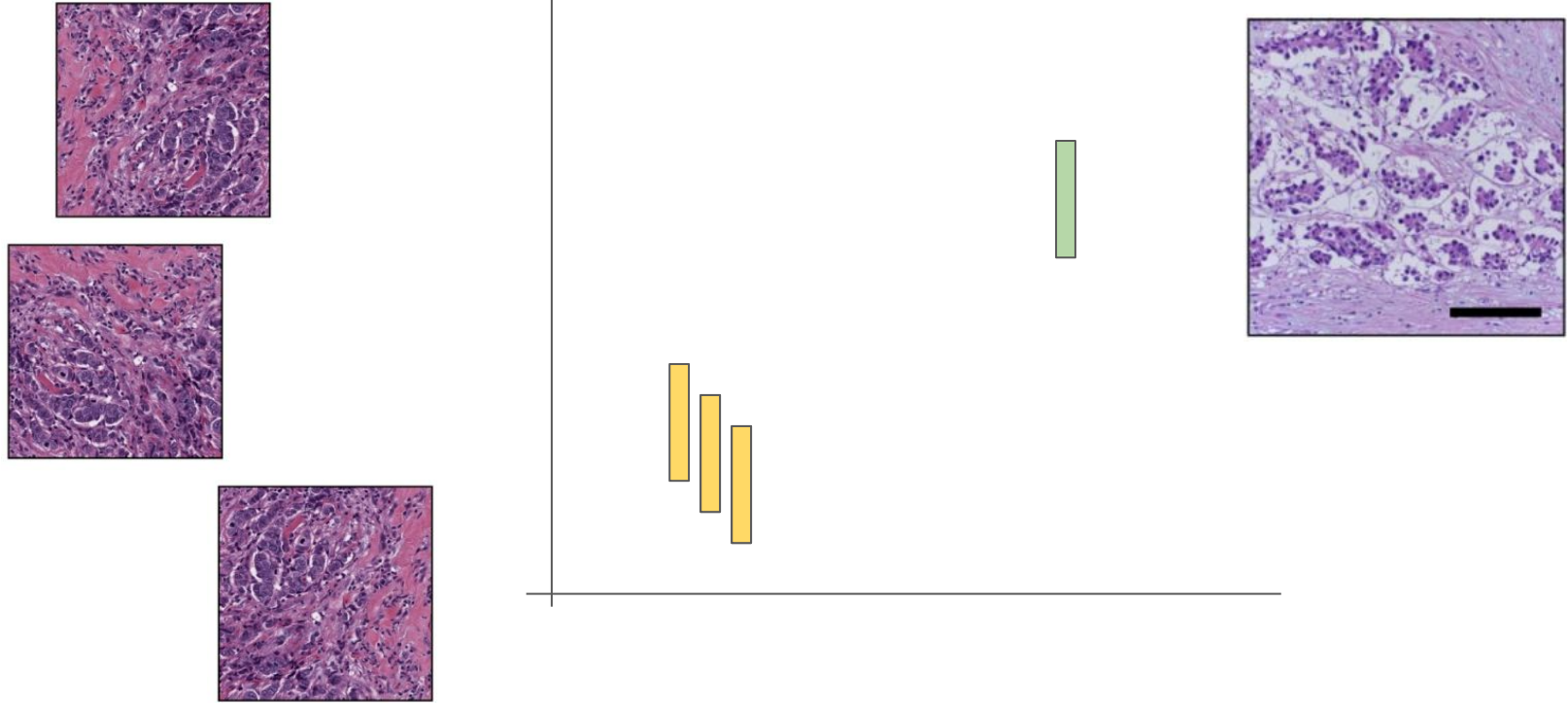
Lu et al., Nat Med 2024

Foundation models

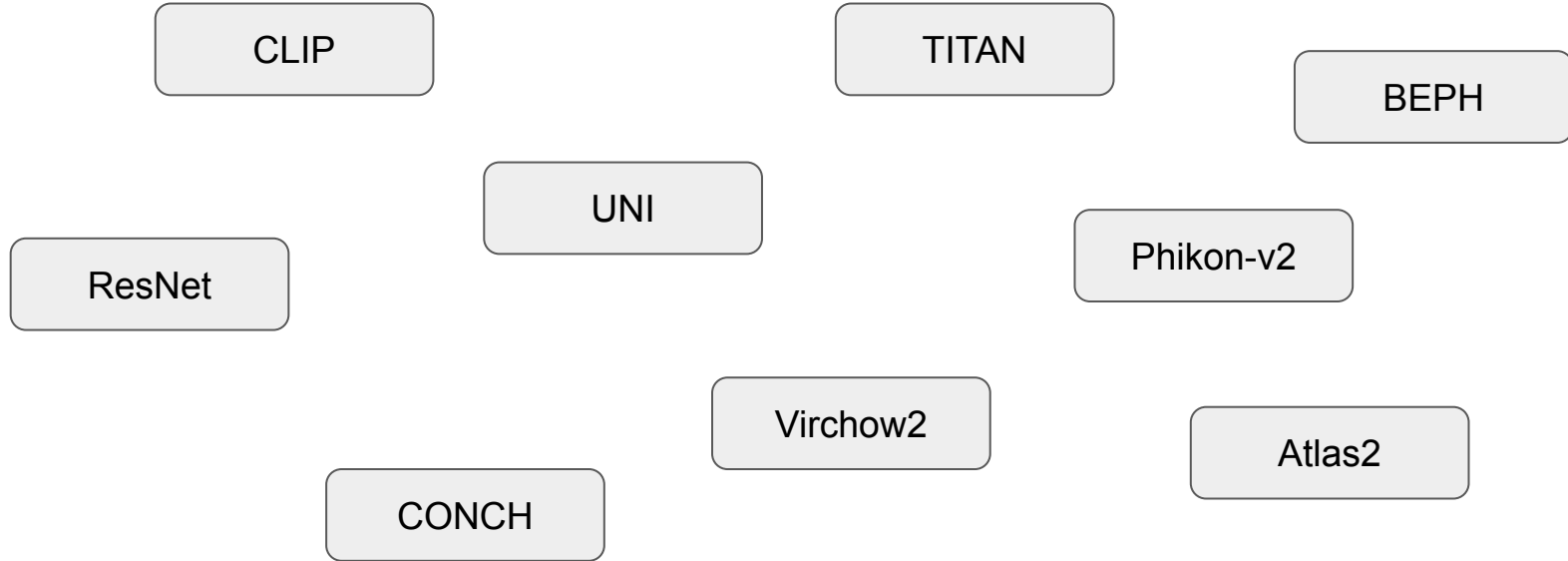


Chen et al., Nat Med 2024

We can also build an atlas



Which embedding is the “best”?



Overall goal

Develop and evaluate an LLM-mediated, retrieval-based system for histopathology patches extracted from WSIs.

Minimal viable product (Phase 0 deliverable):

The MVP should implement a complete end-to-end pipeline that extracts image patches from whole slide images (WSIs), computes feature embeddings for each patch using a pretrained model, and stores these embeddings in a vector index for efficient similarity search. Given a query image patch, the system should retrieve the most similar patches from the database.

Assigned to **HISTORAG**:

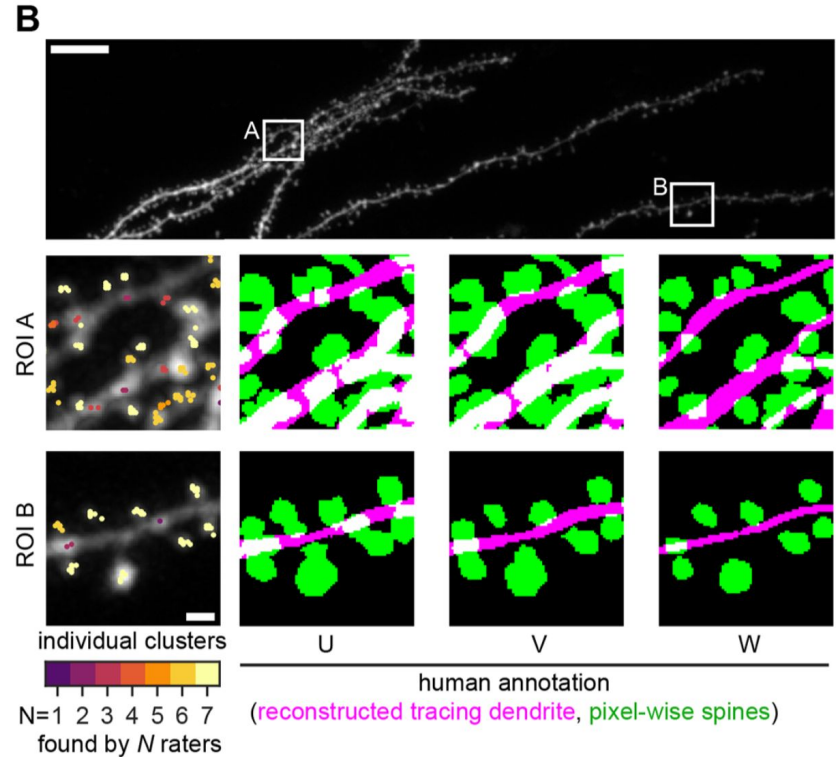
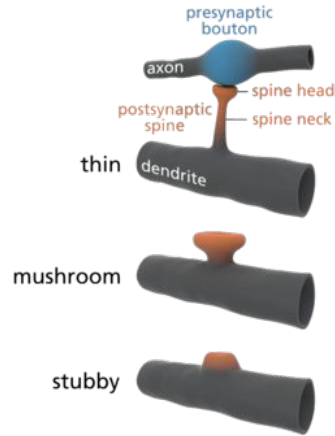
Satyaki Bhattacharjee

Vibhu Gupta

Taimoor Ajmal

NEUROFIND

TL;DR: The project is about developing a Graphical User Interface (GUI) for the re-identification of dendritic spines in microscopic image stacks acquired at different timepoints



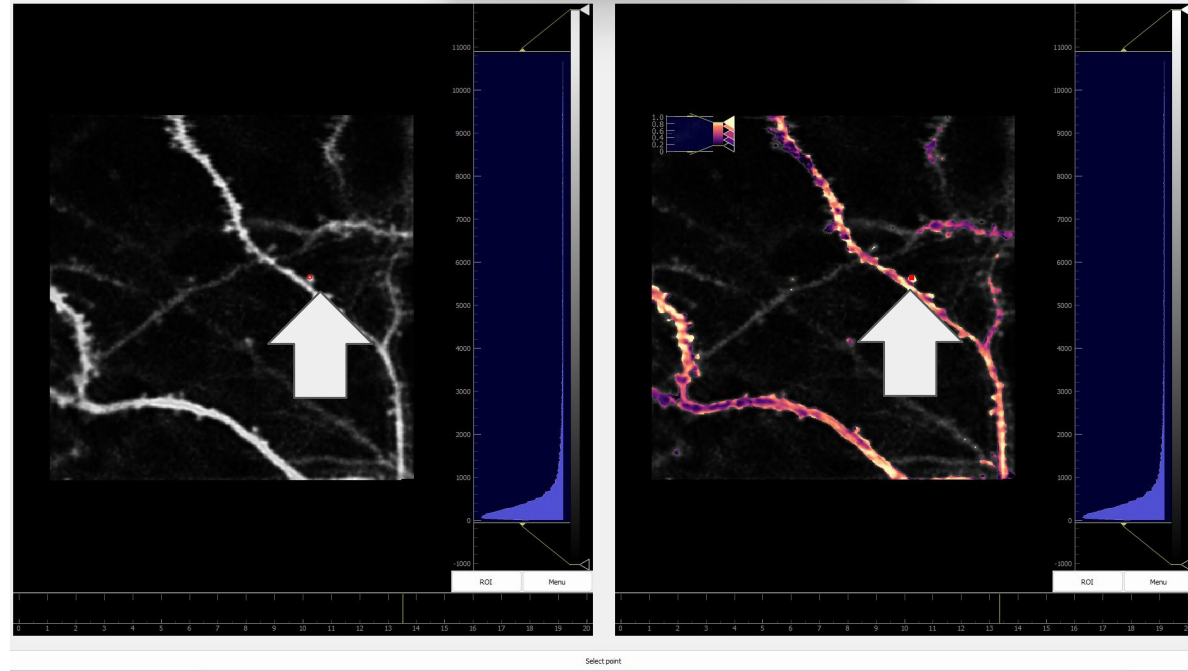
Fernholz et al., 2024, PLOS Computational Biology

Starting point: Vibe-coded GUI

Features:

1. Load and show two image stacks (tif)
2. Select one point in first image stack
3. Show similarities and best matching point in second image stack

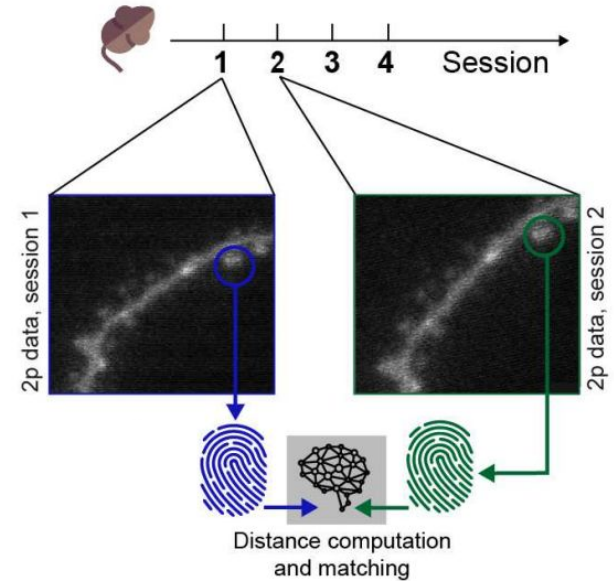
→ Re-create as a napari plugin and improve GUI



Matching principle

- Models are trained in a self-supervised manner to create an embedding of each desired point
→ 'fingerprint'
- Best matches from different imaging session should be from the same neuroanatomic structure, i.e. the same spine

→ **Fine-tuning of models necessary**



Overall goal

The goal of the project is to develop a GUI as a Napari plugin that uses (foundation) models to track dendritic spines across time in consecutive image stacks.

Minimal viable product (Phase 0 deliverable):

The MVP is a napari plugin that provides at least the same functionality as the vibe-coded GUI that already exists. This includes loading two image stacks, selecting a structure in the first image stack and showing the similarities (predicted by one model) with this structure in the second image stack and the best match with the selected structure.

Assigned to **NEUROFIND**:

Arbaz Khan

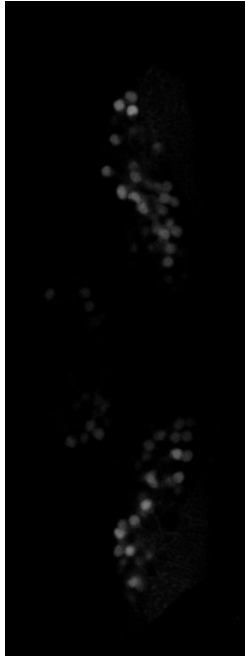
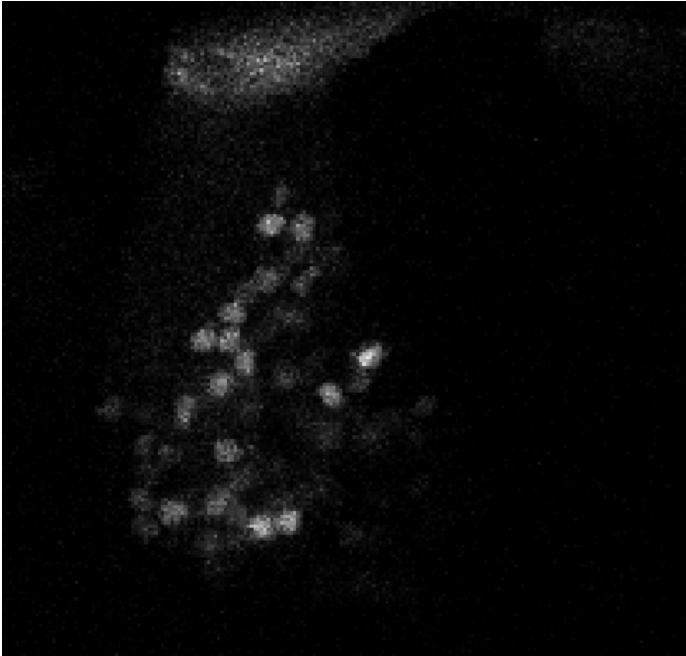
Yulong Wang

Ahmed Sofan

TL;DR: Develop a neuron segmentation pipeline using semantic segmentation for active neurons in microscopic Ca^{2+} imaging data (2D+t). Fluorescent calcium imaging helps understand brain functions by recording neuronal activities. However, manual annotation is extremely time-consuming and subjective. Therefore, an automatic tool should be developed.

We have data from zebrafish, drosophila and mouse

Zebrafish examples:



Overall goal

The goal of this project is to develop a pipeline that automatically segments active neurons in calcium imaging data.

Minimal viable product (Phase 0 deliverable):

The MVP should be a basic segmentation pipeline that runs end-to-end. The pipeline needs to include loading the data, preprocessing, creating a basic semantic segmentation method, and plotting individual neural activity traces.

Assigned to **NEUROSEG**:

Muhammad Shaizal Yasin

Mohammad Karbalaee

Nilamathi Rathinapandi

TL;DR: This project focuses on continuing the development of a graphical user interface (GUI) for motion correction in biomedical image sequences. The current system estimates deformation fields by tracking feature points across frames using the CoTracker3 algorithm.

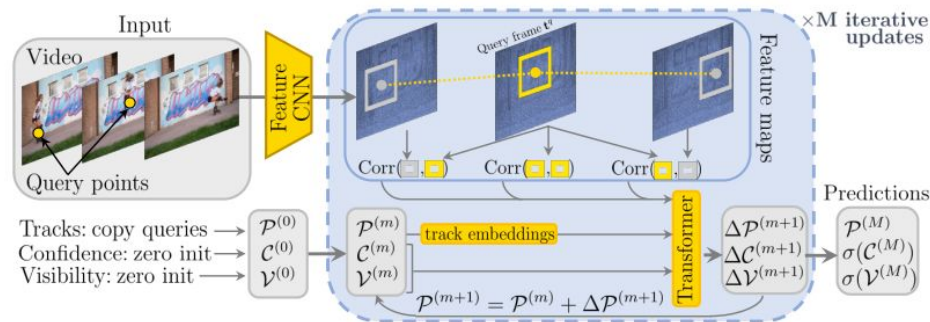
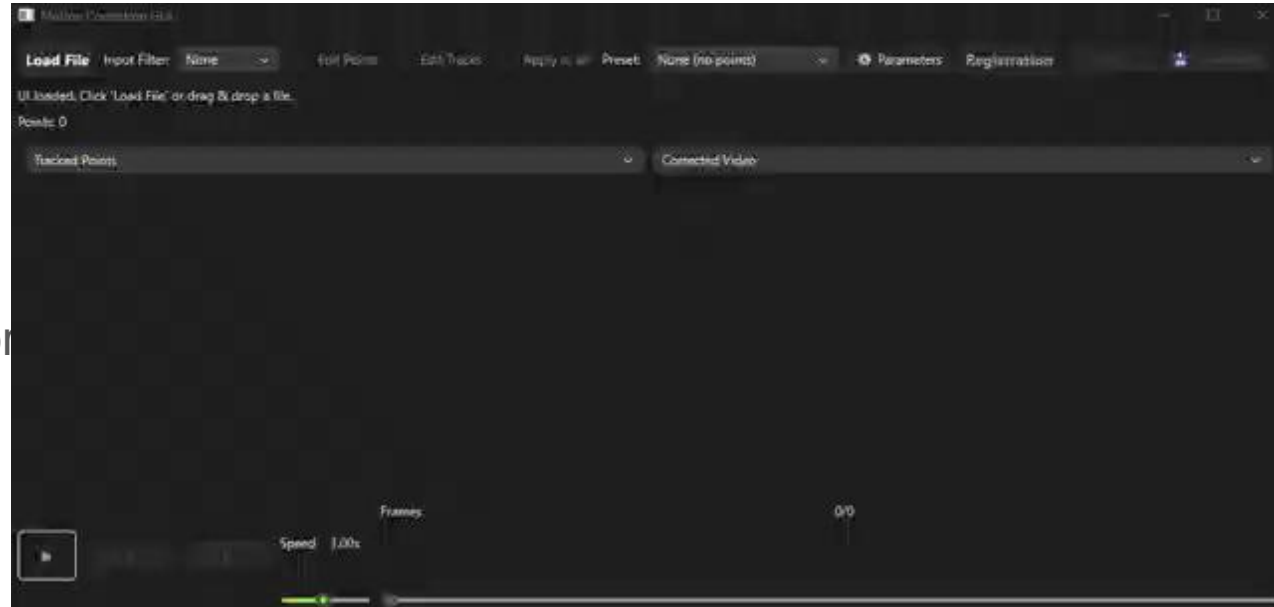


Figure 2: **Architecture.** We compute convolutional features for every frame of the given video, and then the correlations between the feature sampled around the query frame for the query point and all the other frames. We then iteratively update tracks $\mathcal{P}^{(m)} = \mathcal{P}^{(m)} + \Delta \mathcal{P}^{(m+1)}$, confidence $\mathcal{C}^{(m)}$, and visibility $\mathcal{V}^{(m)}$ with a transformer that takes the previous estimates $\mathcal{P}^{(m)}$, $\mathcal{C}^{(m)}$, $\mathcal{V}^{(m)}$ as input.

<https://arxiv.org/pdf/2410.11831>

Current GUI

- Load data
- Track points with CoTracker
- Use displacement field for deformation field
- Warp to compensate for motion



Can we enhance motion compensation by finetuning Cotracker?

- Review GUI and baselines
- Data preparation for finetuning CoTracker (VFSS data, 2P data)
- Flexible/sophisticated tracking on non-starting frames

Overall goal

The goal of this project is to improve our tracking-based motion correction pipeline by extending the GUI and evaluating domain-specific fine-tuning of CoTracker3 regarding tracking performance.

Minimal viable product (Phase 0 deliverable):

By the end of Phase 0, you should run CoTracker3 on biomedical video data and have prepared the datasets to finetune CoTracker on.

Assigned to **CORRECT**:

Aditya Mallik

Mohamed Noaman

TL;DR: CoTracker3 predicts trajectories of points in image sequences (2D+t) but does not directly support volumetric time-series data (3D+t). However, many biomedical imaging modalities produce exactly such datasets (e.g., 3D microscopy time series, CT or MRI sequences). Tracking points in such datasets would enable sophisticated analysis of motion and temporal deformation.

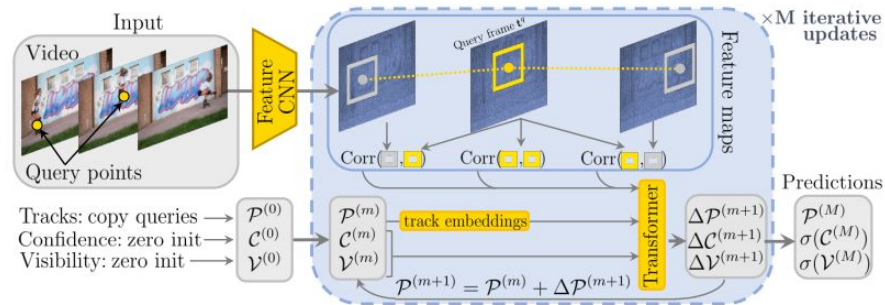


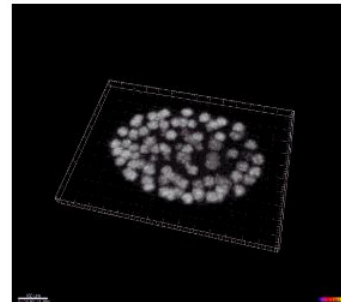
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<https://arxiv.org/pdf/2410.11831>

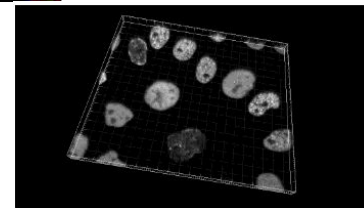
Datasets to be considered

- Cell Tracking Challenge (CE, CHO)
- Multi-breath 4DCT images of lung cancer patients
- [Spreadsheet with 4D Datasets](#)

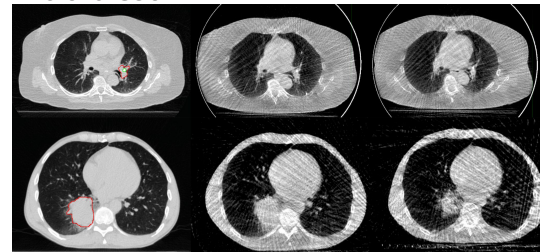
CE



CHO



Multi-breath



Can we track in 3D+t

The CoTracker3, which we use for motion compensation is implemented for 2D+t, but how about tracking in 3D+t?

- Prepare Data & Loading
- Adjust CoTracker & Pipeline
- Visualize and quantifying resulting trajectories

Overall goal

The goal of this project is to Investigate how CoTracker3 can be adapted from 2D+t tracking of 3D+t tracking by enabling the prediction of 3D trajectories over time.

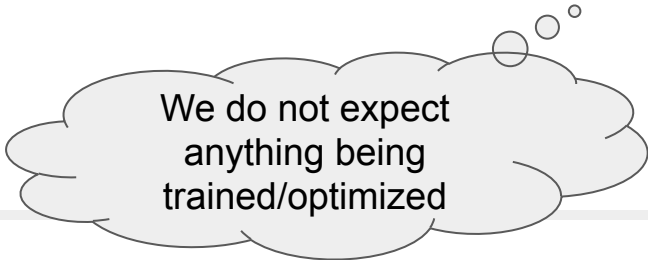
Minimal viable product (Phase 0 deliverable):

By the end of Phase 0, the prototype should “track” points in 3D+t, including preparing one dataset and adjusting the CoTracker architecture to accept and predict 3D tracks.

Assigned to **VOLTRACK**:

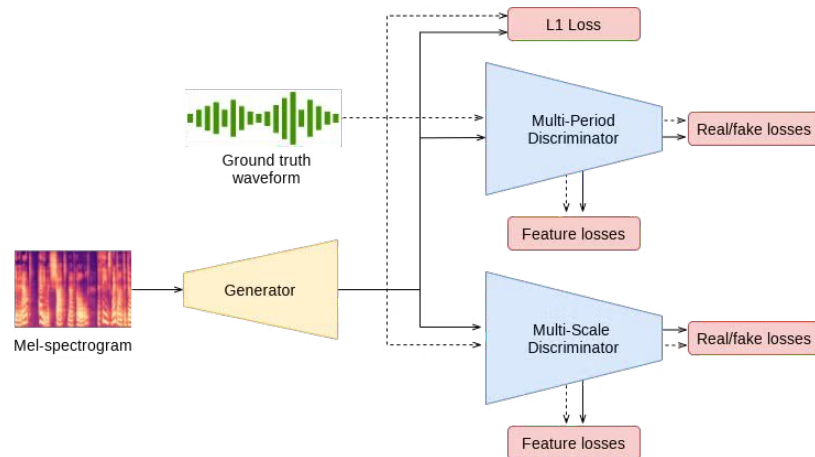
Lucas Bruder

Vivek Kumar Bhagat



We do not expect
anything being
trained/optimized

TL;DR: This project explores **small-scale neural vocoders** to find the sweet spot between **audio quality** and **hardware efficiency**. Students will build and test lightweight models to see how fast and clear they can make speech synthesis on standard computers.



- Baseline Hifi-GAN (<https://github.com/jik876/hifi-gan>): Kong, Jungil, Jaehyeon Kim, and Jaekyoung Bae. "Hifi-gan: Generative adversarial networks for efficient and high fidelity speech synthesis." Advances in neural information processing systems 33 (2020): 17022-17033.
- VOCOS (<https://github.com/gemelo-ai/vocos>): Siuzdak, Hubert. "Vocos: Closing the gap between time-domain and Fourier-based neural vocoders for high-quality audio synthesis." The Twelfth International Conference on Learning Representations.
- LPCNet (<https://github.com/xiph/lpcnet>): Valin, Jean-Marc, and Jan Skoglund. "LPCNet: Improving neural speech synthesis through linear prediction." ICASSP 2019-2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). IEEE, 2019.
- RNDVoC (<https://github.com/Andong-Li-speech/RNDVoC>): Li, Andong, et al. "Learning neural vocoder from range-null space decomposition." Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence. 2025.

Use one dataset to train all vocoder models (f.e. LibriSpeech) and evaluate audio quality (Mel Cepstral Distortion / spectral metrics) and efficiency (parameter count, model size, inference time)

- How far can model size be reduced before quality drops noticeably?
- Does quantization/pruning help enough to justify the quality loss?
- Which model is most suitable for real-time CPU synthesis?

Overall goal

Systematically compare efficient neural vocoders to determine the optimal balance between model complexity, generation speed, and perceptual audio quality.

Minimal viable product (Phase 0 deliverable):

- Data Pipeline: Dataset selection and feature preprocessing.
- Baseline: Training a standard lightweight vocoder and evaluation of initial audio quality and runtime metrics.

Assigned to **VOCODE**:

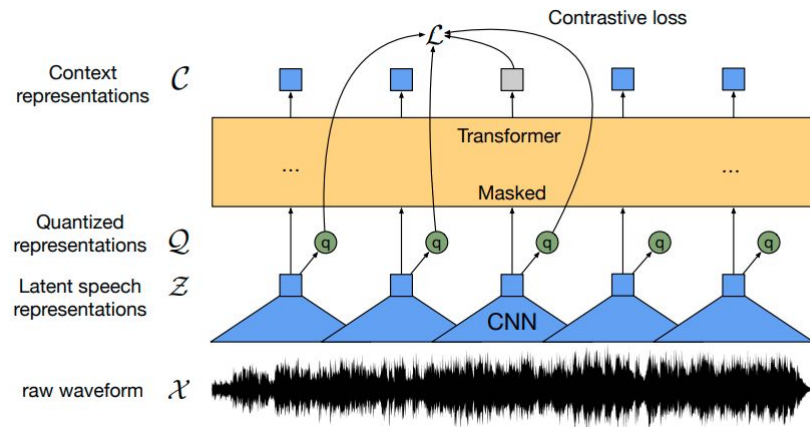
Rahul Sawhney

Jinying Chen

Uddipan Basu Bir

FEATURE

This project evaluates how wav2vec models represent phonological features (like voicing and articulation) across different languages. By comparing multilingual and cross-lingual pretraining, the study explores whether these speech properties are encoded universally and how fine-tuning can better adapt these representations.



- Pretrained Model comparison: wav2vec 2.0 base vs XLSR / multilingual wav2vec 2.0
 - <https://huggingface.co/facebook/wav2vec2-base>
 - <https://huggingface.co/facebook/wav2vec2-large-xlsr-53>
 - <https://huggingface.co/facebook/wav2vec2-xls-r-300>
- Dataset:
 - FLEURS (<https://huggingface.co/datasets/google/fleurs>): Goyal, Naman, et al. "The Flores-101 evaluation benchmark for low-resource and multilingual machine translation." Transactions of the Association for Computational Linguistics 10 (2022): 522-538.

What can we analyze?

- Which pretrained model best captures language-independent phonological features?
- Are multilingual models better than monolingual ones for cross-lingual phonological probing?
- Which layers encode phonological information most strongly?
- Which phonological features transfer well across languages, and which are language-specific?

Overall goal

The goal of this project is to systematically compare different wav2vec / wav2vec 2.0 model variants and evaluate how well phonological features are represented across different languages and if certain languages share latent phonological features.

Minimal viable product (Phase 0 deliverable):

By the end of Phase 0, the project should produce a working end-to-end prototype. This includes selecting several pretrained wav2vec-family models, choosing a multilingual speech dataset with phoneme or phonological feature labels, extracting model representations, training simple probing classifiers, and evaluating how well different phonological features can be predicted across languages.

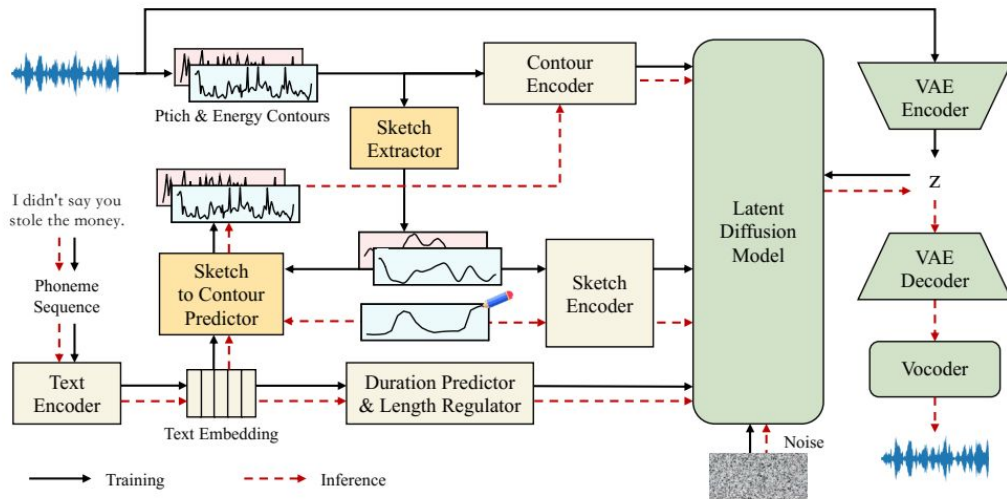
Assigned to **FEATURE**:

Muhammad Muzzamil Anwaar Chaudhry

~~Maitreya Mohapatra~~

Muhammad Haider Zaidi

This project enhances the DrawSpeech (Paper + Code: <https://happycolor.github.io/DrawSpeech/>) framework by shifting from word-level to phoneme-level precision, allowing users to "sketch" expressive controls for individual speech sounds while expanding the system's capabilities to include manual adjustments for pitch, rhythm, and timbre.



- Does phoneme-level sketch conditioning provide better fine-grained control than word-level conditioning?
- Which speech attribute is most feasible to control at phoneme level: pitch, energy, or rhythm?
- How much control precision can be added before naturalness degrades?
- Are phoneme-level controls stable and interpretable for users?

- Baseline: original DrawSpeech-inspired setup
- Main change: phoneme-level sketch conditioning
- Train on LibriSpeech dataset and explore the possibility to train on expressive speech datasets
- Focus: prosody / pitch control only
- Output: controlled samples + objective comparison + small listening test

Overall goal

The goal of this project is to adapt DrawSpeech-style sketch conditioning to phoneme-level speech synthesis control and systematically evaluate which control dimensions work best: prosody only, or extended control over pitch, rhythm, and timbre.

Minimal viable product (Phase 0 deliverable):

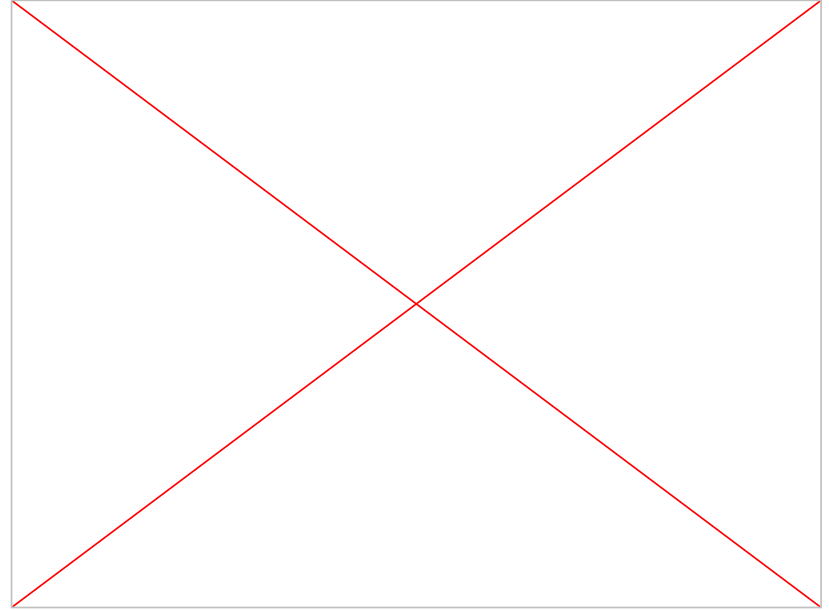
By the end of Phase 0, the project should produce a working end-to-end prototype. This includes reproducing a simplified DrawSpeech-style baseline, replacing word-level sketch control with phoneme-level alignment and conditioning, training the model on a suitable expressive speech dataset, and generating controlled speech samples.

Assigned to **DRAW**:

Neelima Kawatra

Amro Gamea

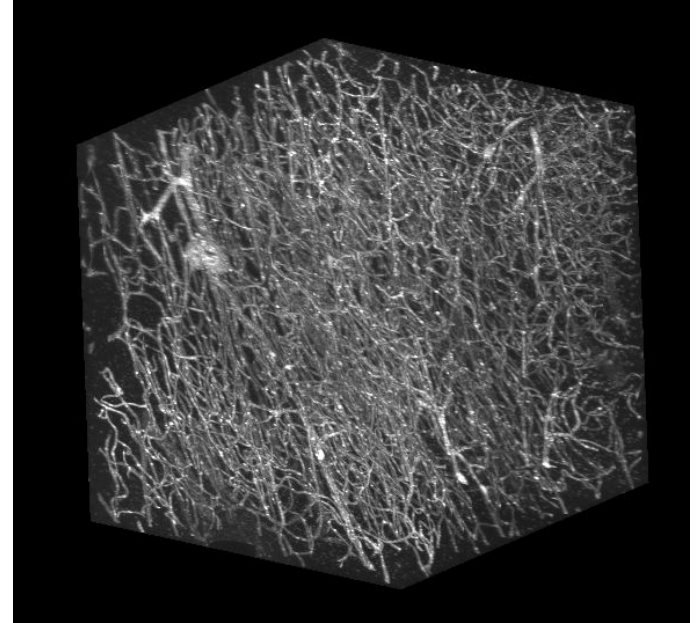
TL;DR: This project builds a small pipeline to analyze and simulate blood flow in a vascular network extracted from a 3D mesoscale microscopy stack. The vessels are represented as a graph derived from segmentation and skeletonization.



Datasets to be considered

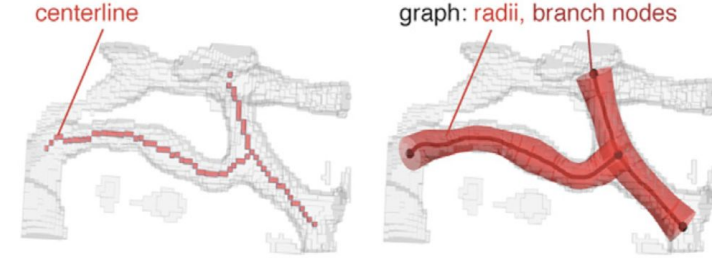
Not published mesoscale human vascular
microscopy data

Small stack, middle, whole sample



What can we investigate?

Graph representations allow efficient analysis of network topology and enable simplified blood flow simulations. Graph traversal algorithms such as Breadth-First Search (BFS) and Depth-First Search (DFS) can be used to explore connectivity, hierarchy, and flow paths within the vascular network.



Kirst, C., Skriabine, S., Vieites-Prado, A., Topilko, T., Bertin, P., Gerschenfeld, G., Verny, F., Topilko, P., Michalski, N., Tessier-Lavigne, M., & Renier, N. (2020). Mapping the Fine-Scale Organization and Plasticity of the Brain Vasculature. *Cell*, 180(4), 780–795.e25. <https://doi.org/10.1016/j.cell.2020.01.028>

Overall goal

The goal of this project is to investigate a graph-based pipeline for analyzing and simulating blood flow in a vascular network.

Minimal viable product (Phase 0 deliverable):

End-to-end prototype:

- Load vascular graph data, segmentation and raw data
- Visualize the network in Napari
- Baseline for blood flow simulation: Breadth-First Search
- Render small stack

Assigned to **FLOW**:

Yifei Li

Dipesh Mann